

In the previous page, we introduced exponential generating functions and moment generating functions. Here we will compute the exponential generating function for the Bell numbers and the moment generating function for $X \sim Poi(\lambda)$. We will observe **Dobiński's formula**.

The **Bell numbers** B_n count the number of ways to partition n elements of a set into nonempty subsets. They have the following recursion for $n \geq 1$

$$B_n = \sum_{i=1}^n \binom{n-1}{i-1} B_{n-i}$$

which we can show with the method of picking a distinguished element $\star$. We will sum over all the possible sizes k of a subset containing $\star$. In order to pick the remaining elements of the subset we have $\binom{n-1}{k-1}$ options. Then we need to partition the rest of the set into nonempty subsets, of which there are B_{n-k} ways to do so.

We can write the exponential generating function for these numbers as

$$G_{B_n}(x) = \sum_{n \geq 0} B_n \frac{x^n}{n!}.$$

If we take the derivative of this it puts it in a form that is useful for our recurrence

$$\frac{d}{dx} G_{B_n}(x) = \sum_{n \geq 0} B_n \frac{x^{n-1}}{(n-1)!} = \sum_{n \geq 0} \left[\sum_{k=1}^n \binom{n-1}{k-1} B_{n-k} \right] \frac{x^{n-1}}{(n-1)!}$$

If we cancel the $(n-1)!$ in the top and bottom we can separate the terms to write them as

$$\frac{d}{dx} G_{B_n}(x) = \sum_{n \geq 0} \left[\sum_{k=1}^n \frac{x^{k-1}}{(k-1)!} B_{n-k} \frac{x^{n-k}}{(n-k)!} \right]$$

Since the sum of our exponents $k-1+n-k$ is constant for a fixed n and the pair $(k-1, n-k)$ goes from $(0, n-1)$ to $(n-1, 0)$ as k varies, all pairs of non negative integer values are attained for the exponents of our variable and we can reindex this sum as

$$\frac{d}{dx} G_{B_n}(x) = \sum_{i \geq 0} \sum_{j \geq 0} \frac{x^i}{i!} B_{n-k} \frac{x^j}{j!} = \sum_{i \geq 0} \frac{x^i}{i!} \sum_{j \geq 0} B_j \frac{x^j}{j!} = e^x G_{B_n}(x).$$

So we have the differential equation $G'_{B_n}(x) = e^x G_{B_n}(x)$ with initial condition $G_{B_n}(0) = 1$. Using the method of separation of variables we get that $G_{B_n}(x) = e^{e^x - 1}$.

Now we will find the moment generating function of $X \sim Poi(\lambda)$. Given that $\mathbb{P}[X = k] = \frac{\lambda^k e^{-\lambda}}{k!}$ then we have that

$$\mathbb{E}[e^{tX}] = \sum_{n \geq 0} \frac{\lambda^n e^{-\lambda}}{n!} e^{tn} = e^{-\lambda} \sum_{n \geq 0} \frac{(\lambda e^t)^n}{n!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

So we can see that the n^{th} moment of $Poi(1) = B_n$.